

Que1. Create the following transfer function using `tf(num,den)` command. $H(s) = (s + 5)/(s^2 + 8s + 16)$ and then convert from tf model (polynomial form) to zpk model (factored form) using MATLAB

Solution:- Code:

```
num = [1 5];      % Numerator coefficients (s + 5)
den = [1 8 16];   % Denominator coefficients (s^2 + 8s + 16)
sys = tf(num, den); % Create transfer function [cite: 12, 39]
sys_zpk = zpk(sys); % Convert to zpk model [cite: 12, 39]
disp('Transfer Function H(s):');
printsys(num, den);
disp('ZPK Model:');
disp(sys_zpk);
```

Output:

Transfer Function H(s):

num/den =

$$\frac{s + 5}{s^2 + 8s + 16}$$

ZPK Model:

[zpk](#) with properties:

Z: {-5}

P: {[2×1 double]}

K: 1

DisplayFormat: 'roots'

Variable: 's'

IODelay: 0

InputDelay: 0

OutputDelay: 0

InputName: {''}

InputUnit: {''}

InputGroup: [1×1 struct]

OutputName: {''}

OutputUnit: {''}

OutputGroup: [1×1 struct]

Notes: [0×1 string]

UserData: []

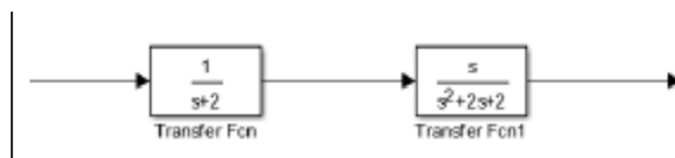
Name: ''

Ts: 0

TimeUnit: 'seconds'

SamplingGrid: [1×1 struct]

Que2. Two systems with transfer functions are mentioned $G_1(s) = 1/(s+2)$ and $G_2(s) = s/(s^2+2s+2)$
Find the transfer function of the system defined by the two cascaded subsystems using MATLAB.

**Solution:**

% Q2: Cascaded (Series) systems [cite: 21]

sys1 = tf(1, [1 2]); % $G_1(s) = 1/(s+2)$ sys2 = tf([1 0], [1 2 2]); % $G_2(s) = s/(s^2+2s+2)$

sys3 = series(sys1, sys2); % Calculate cascaded TF [cite: 21, 46]

disp('Transfer Function of Cascaded System (G1 * G2):');

sys3

Output

Transfer Function of Cascaded System ($G1 * G2$):

sys3 =

$$\frac{s}{s^3 + 4s^2 + 6s + 4}$$

Continuous-time transfer function.

Que.3 Two systems with transfer functions are given $G1(s) = 1/(s+2)$ and $G2(s) = s/(s^2+2s+2)$
Find the transfer function of the system defined by the two subsystems with parallel connection using MATLAB.

Solution:

```
% Q3: Parallel connection [cite: 22]
sys1 = tf(1, [1 2]);
sys2 = tf([1 0], [1 2 2]);
sys_para = parallel(sys1, sys2); % Calculate parallel TF [cite: 22, 54]
disp('Transfer Function of Parallel System (G1 + G2):');
sys_para
```

Output:

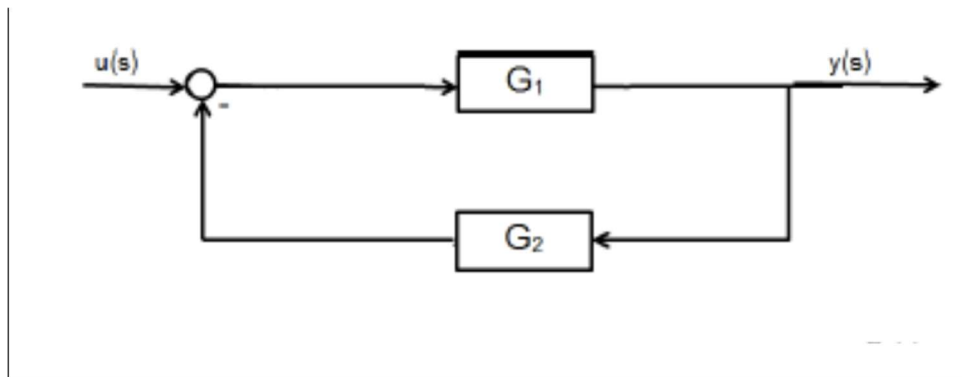
Transfer Function of Parallel System ($G1 + G2$):

sys_para =

$$\frac{2s^2 + 4s + 2}{s^3 + 4s^2 + 6s + 4}$$

Continuous-time transfer function.

Que4. Given two systems with transfer functions $G_1(s) = 1/(s+2)$ and $G_2(s) = s/(s^2+2s+2)$
Find the transfer function of the system with feedback elimination rule using MATLAB.



Solution:

% Q4: Feedback elimination [cite: 23]

```
sys1 = tf(1, [1 2]); % G1
```

```
sys2 = tf([1 0], [1 2 2]); % G2 (Feedback path)
```

```
sys_fb = feedback(sys1, sys2); % Feedback elimination rule [cite: 23, 56]
```

```
disp('Transfer Function with Feedback:');
```

```
sys_fb
```

Output:

Transfer Function with Feedback:

sys_fb =

$$\frac{s^2 + 2s + 2}{s^3 + 4s^2 + 7s + 4}$$

Continuous-time transfer function.

Que5. For a given open loop transfer function of a system, plot Root locus using rlocus() command in MATLAB. Put K=48.

$$G(s)H(s)=K/(s^3+6s^2+8s)$$

Solution:

% Q5: Root Locus Plot [cite: 28, 61]

K = 48;

num = [K]; % Numerator is K [cite: 62, 63]

den = [1 6 8 0]; % Denominator: $s(s^2 + 6s + 8) \rightarrow s^3 + 6s^2 + 8s$ [cite: 63]

G = tf(num, den);

figure(1);

rlocus(G); % Generate Root Locus plot [cite: 28, 61]

title('Root Locus for $G(s)H(s) = 48 / (s^3 + 6s^2 + 8s)$ ');

grid on;

Output:

